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Investor Presentation: Global Technology | Asia Pacific

Chipflation

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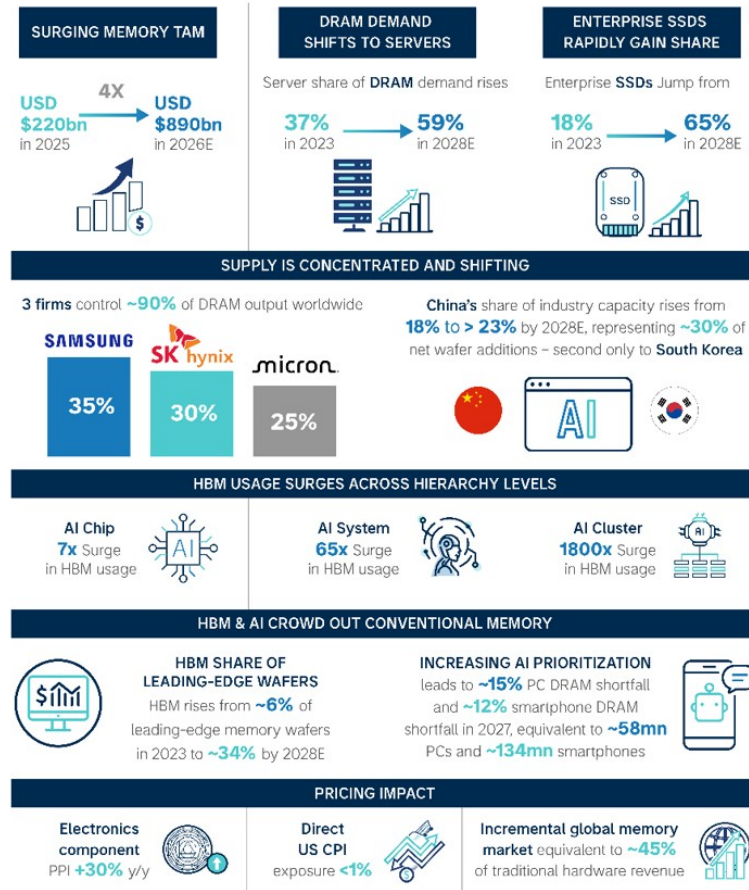


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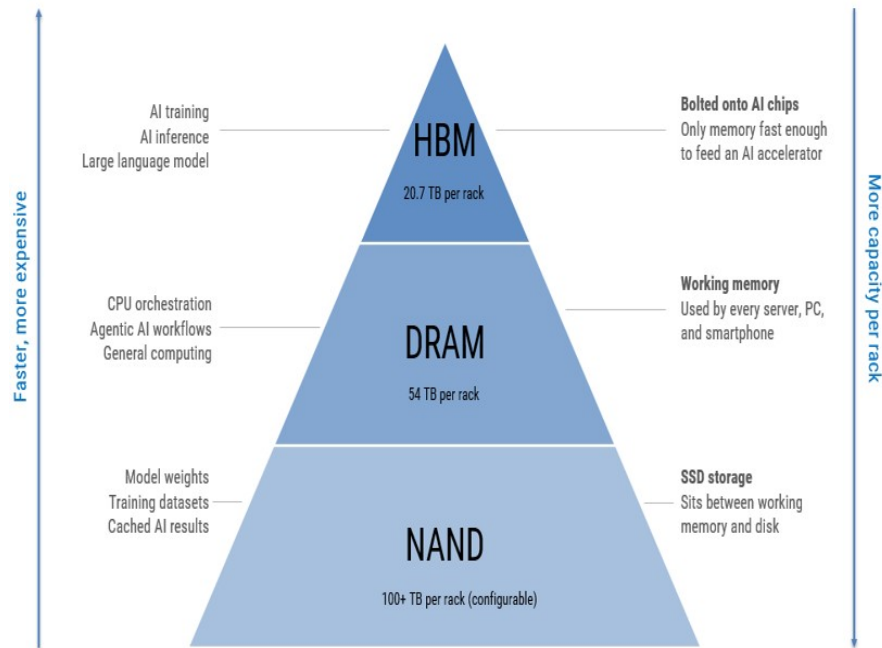
Chipflation – The Story in Numbers



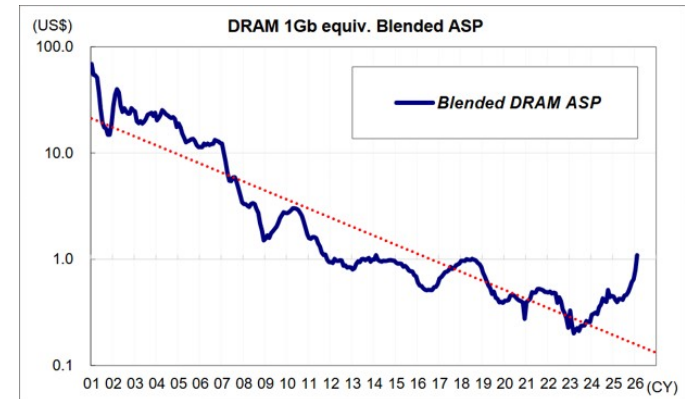
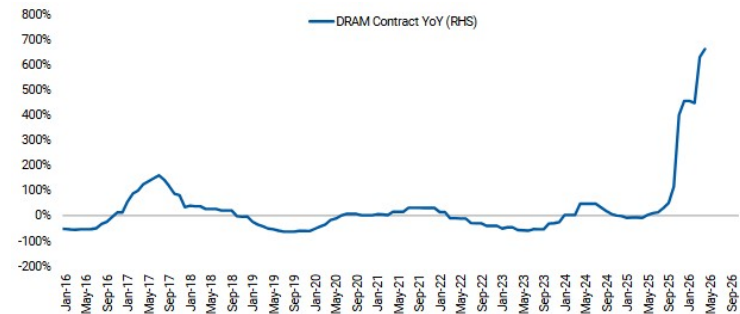
Source: Morgan Stanley Research.

AI is turning memory into a structural bottleneck

- AI servers are becoming memory systems



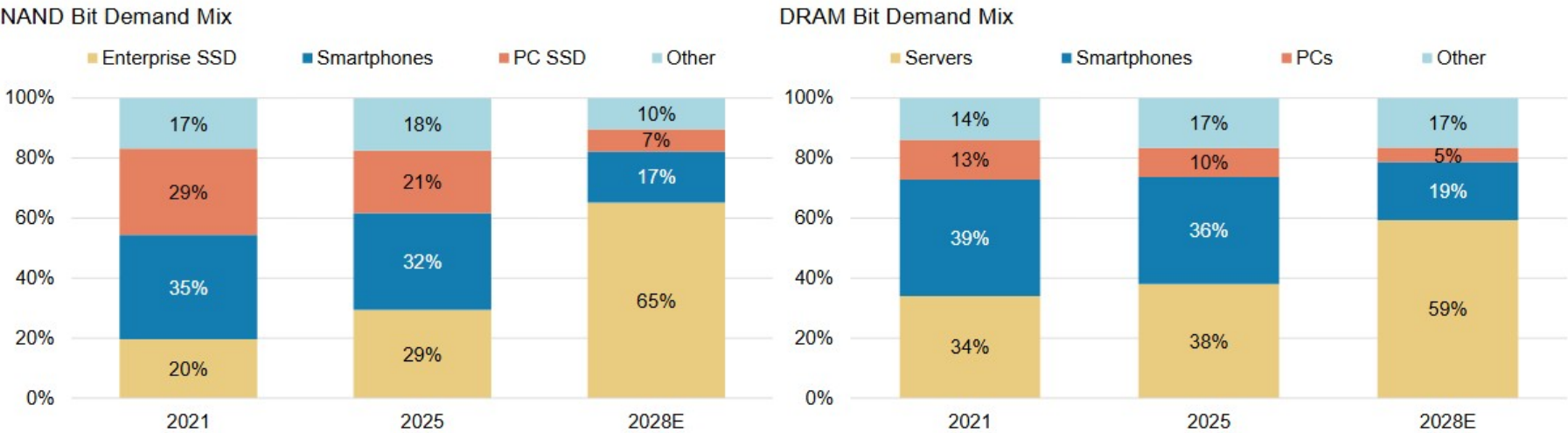
- Prices for memory have risen more than 6-fold over the last year



Source: WSTS, TrendForce, Morgan Stanley Research.

Demand is shifting from consumer electronics to AI/server infrastructure

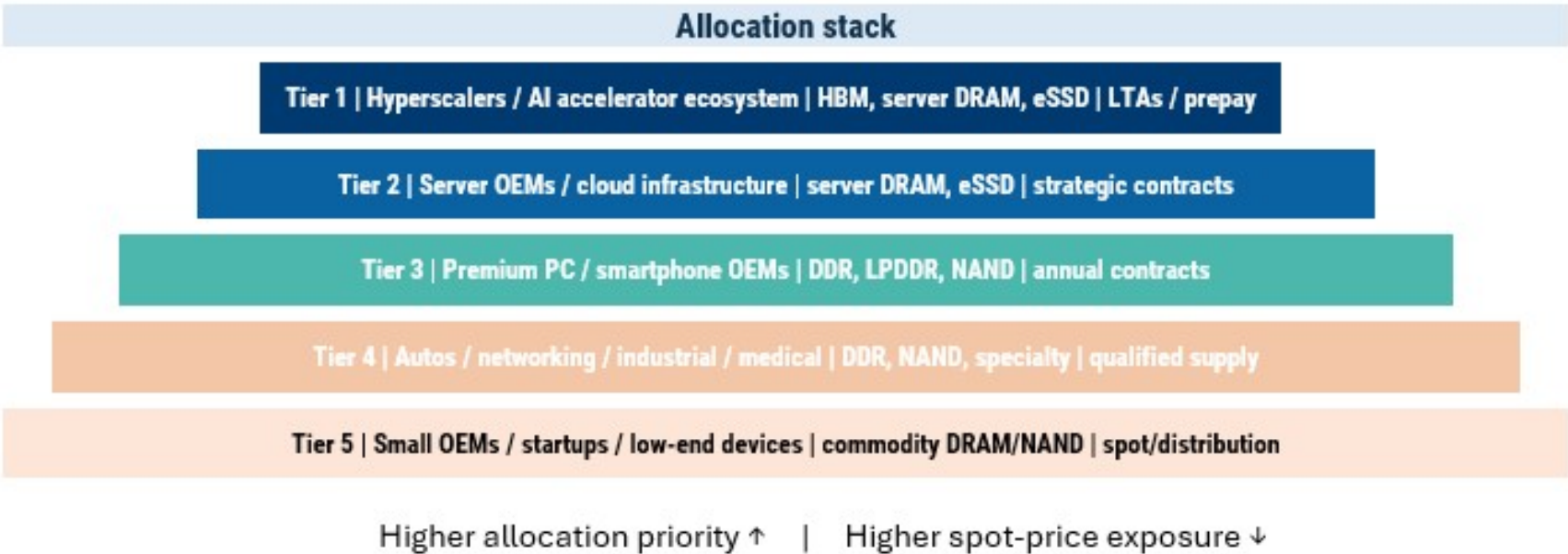
- Memory demand is overwhelmingly shifting towards data center technologies and away from more consumer-facing Smartphone and PC markets



Source: TrendForce, Morgan Stanley Research.

There is a clear divide between memory suppliers and customers

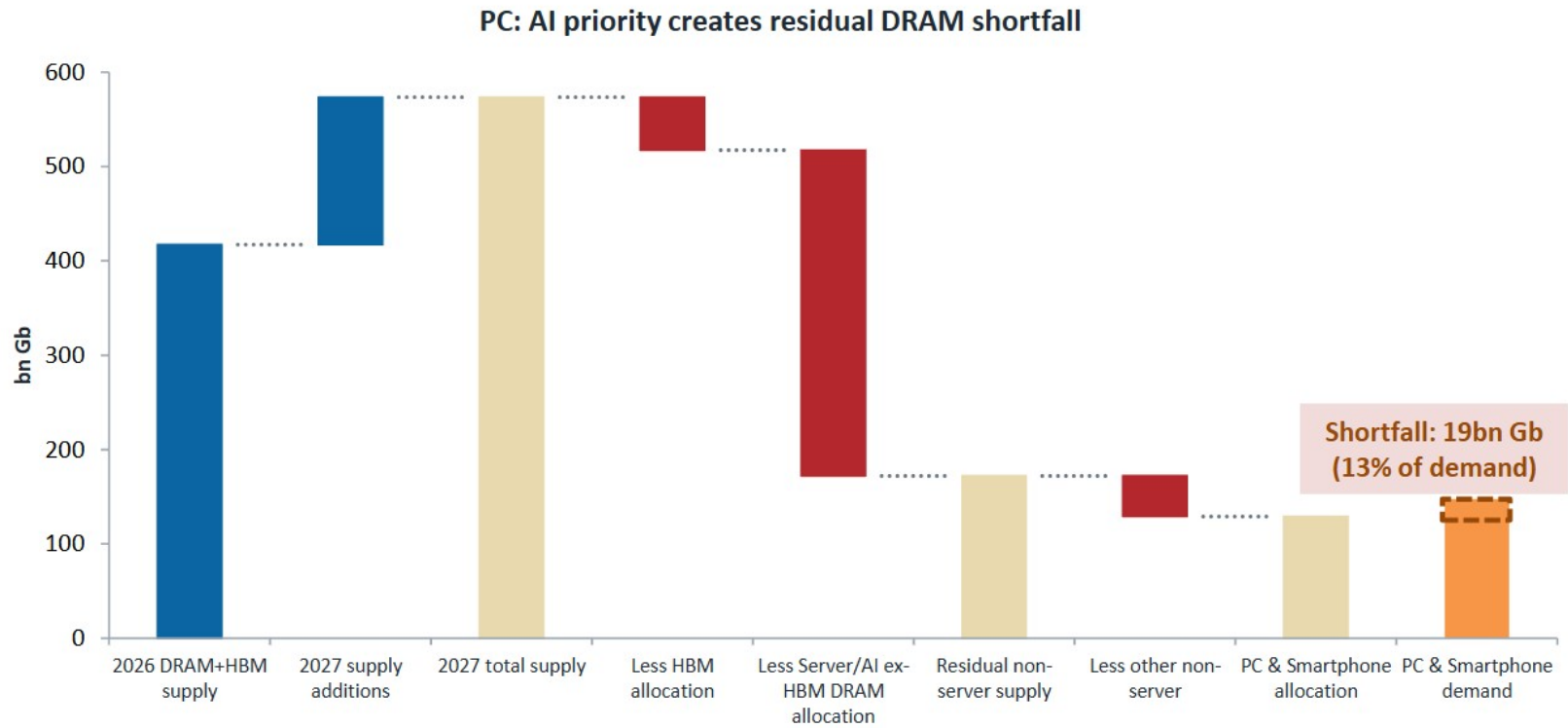
- Buyer Hierarchy: Large cloud buyers can secure supply and capitalize higher costs, while non-AI buyers face higher COGS and weaker allocation, price increases, spec cuts and delayed launches



Source: Morgan Stanley Research.

AI prioritization turns 2027 supply growth into a consumer memory shortfall

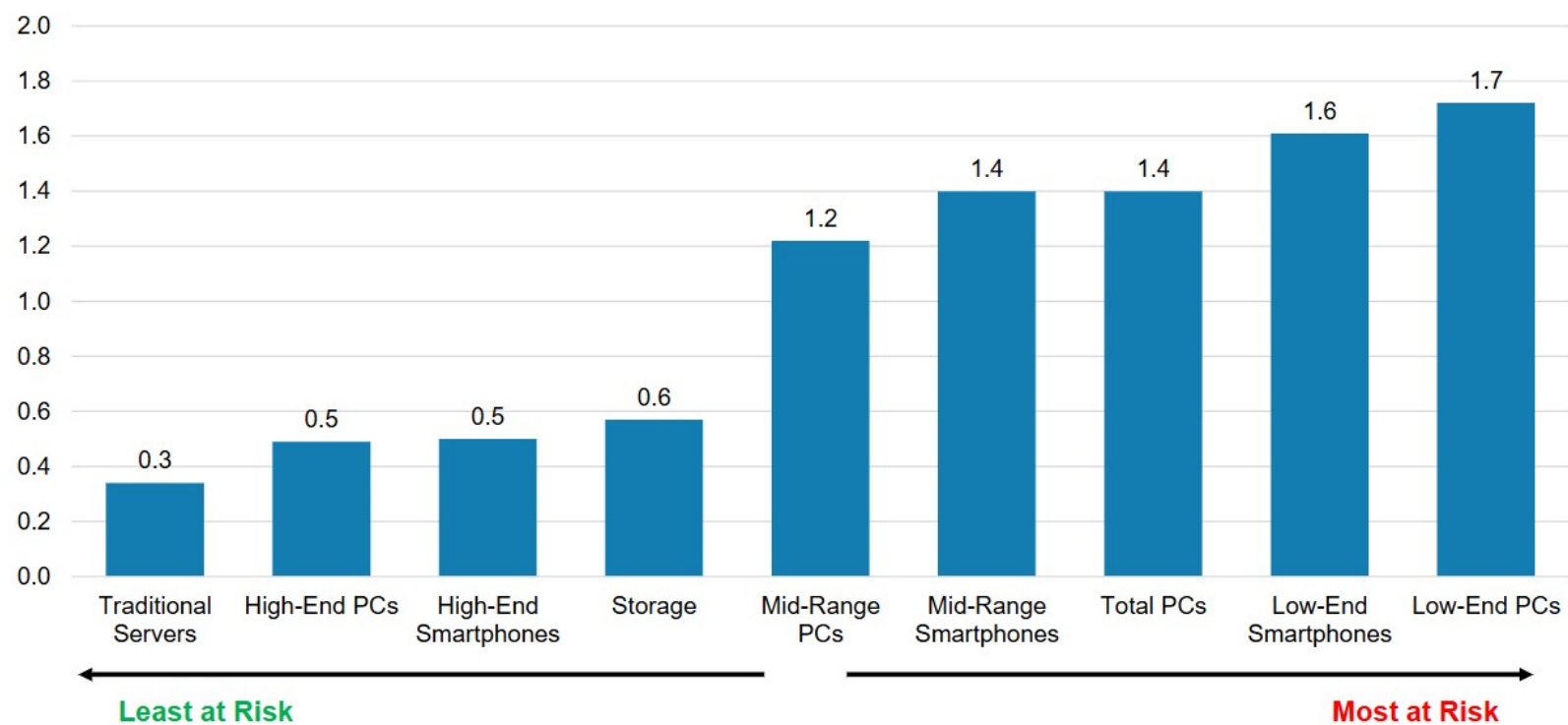
- Our memory sufficiency framework suggests PC and smartphone in aggregate could see 13% memory shortfall



Source: Morgan Stanley Research.

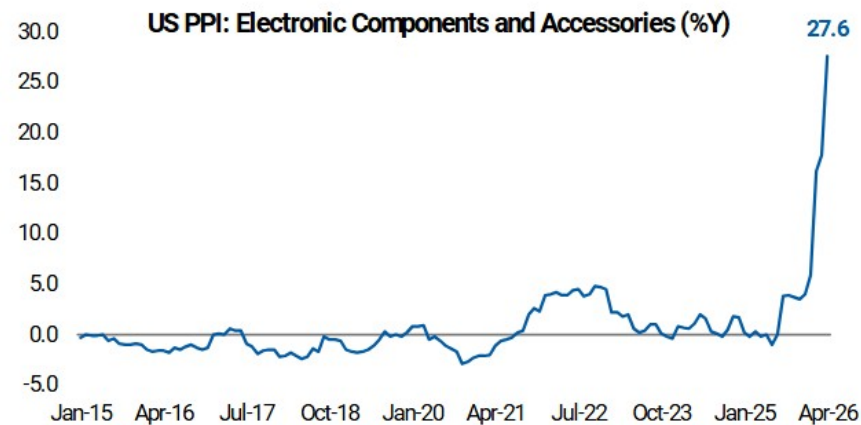
Low-end smartphones and PCs are more exposed to demand destruction

Demand Elasticity by Hardware Product



Source: IDC, Company data, AlphaWise, Morgan Stanley Research. Demand elasticity takes the absolute value. PC data goes back to 1995, for storage it starts in 2008, servers in 2003 and smartphones in 2007.

Chipflation is translating into higher producer price inflation



CPI component	Memory costs effect on CPI in 2026 (pp)	CPI weight
PCs	15	0.30%
Smartphones	15	0.20%
TVs	10	0.10%
Cars	0	3.80%
Major household appliances	5	0.06%
Game consoles	125	0.01%
Impact on headline CPI: PCs and smartphones only	0.08	
Total impact on headline CPI	0.10	

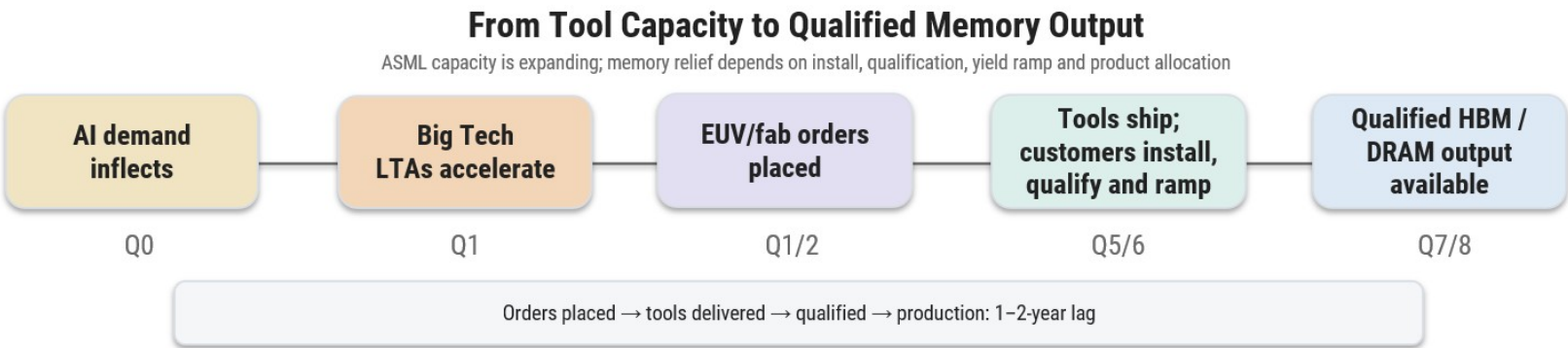
Source: BLS, Morgan Stanley Research forecasts. Note: Smartphones and game consoles weights are estimates.

Policy could ease pressure, but would take years

- Federal support is possible ...

Memory Category	Policy Objective	Policy Tools
HBM/Advanced DRAM	Protect strategic AI inputs & frontier capabilities; delay adversary access	Maintain de-risking (export controls) strategy; expand supply-side efforts to expand trusted capacity
Commodity/legacy memory	Avoid overly restrictive policy; facilitate trusted supply & alleviate cost pressures	Differentiated licensing regimes; targeted domestic/allied capacity support vis-à-vis supply-side initiatives; procurement for critical sectors

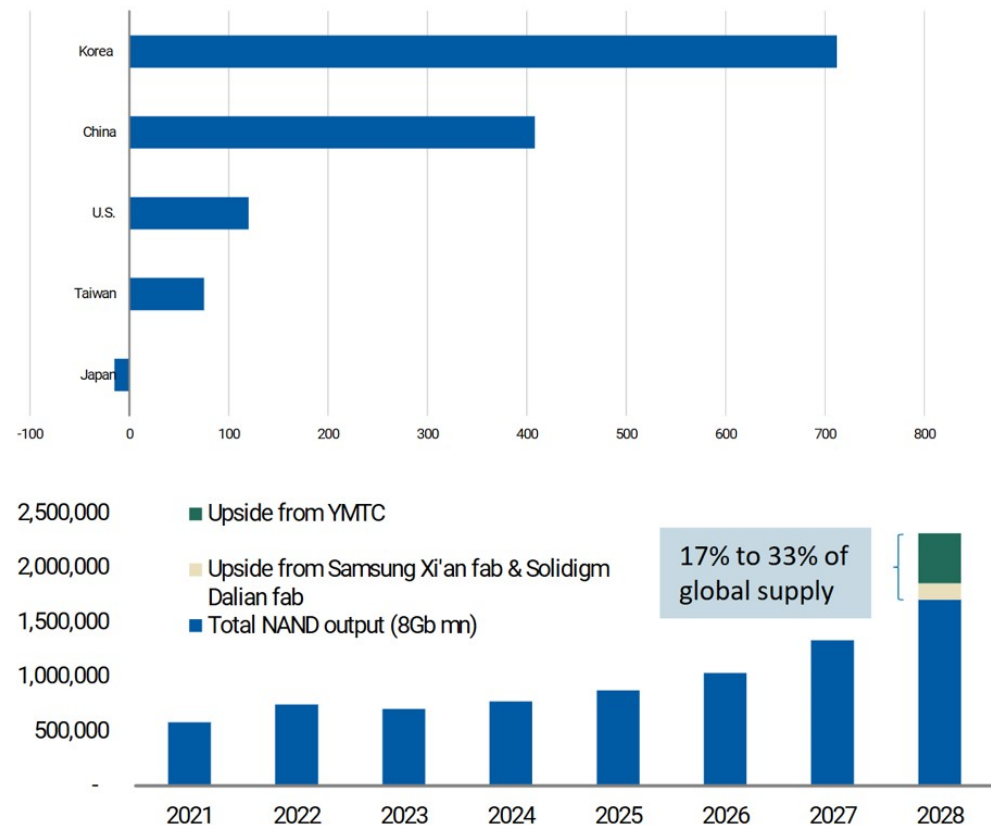
- ... but key constraints exist: procedure, timing & public perception



Source: Morgan Stanley Research.

Could China be the swing factor?

- Mainland China accounts for ~30% of net wafer additions over 2023-28E, second only to South Korea
- However, not all capacity is equally advanced, accessible, or trusted.
- On NAND, the potential uplift is considerable: combining YMTC capacity and yield expansion with node migration at Samsung's Xi'an and Solidigm's Dalian facilities, China-based incremental output could reach 17-33% of 2028E global NAND supply.
- This acceleration scenario assumes a meaningful relaxation of US export controls and is not our base case, as we do not assume the EUV ban on exports to China is lifted.
- The key gating factor is not Chinese ambition or investment appetite, but access to leading-edge equipment.



Source: Morgan Stanley Research.

How to play the theme?

Global Exposure Across the Stack ? Names by exposure					
CPU • NVDA • Intel • AMD • Arm	DRAM • Samsung • Hynix • Micron	NAND • Kioxia • SanDisk	HDD • Seagate • WDC • TDK	FOUNDRY • TSMC	IC-design • GUC • Egis
PCB/Substrate/CCL & Materials • SEMCO • Unimicron • NYPCB • Ibiden • Nittobo • MEC			BMC, CPU & Memory interface • Aspeed • Renesas • Montage • WPG • AP Memory		
MLCC & CPU socket • Murata • TDK • Yageo • FIT Hon Teng • Lotes		ODM • Wiwynn • Hon Hai	SPE • ASML • ASMi • AMAT • Besi • KLAC • Tokyo Electron • Ulvac • Wonik		

Source: Morgan Stanley Research.

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

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